Decoding Quasi-Cyclic codes is NP-complete

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Abstract

This paper establishes the computational complexity of the Quasi-Cyclic Syndrome Decoding Problem (QC-SDP). We introduce a novel characterization of Quasi-Cyclic (QC) codes that generalizes their structure and reveals connections to random codes. Building on this characterization, we prove that QC-SDP is NP-hard and that its decision variant is NP-complete. These results provide the first formal complexity treatment of QC-SDP in its general form, addressing a fundamental open question in code-based cryptography. We also discuss the practical implications and limitations of these theoretical findings for cryptographic applications.

1 Introduction

Quantum computing represents a paradigm shift in computational models, harnessing quantum mechanical phenomena to solve certain problems exponentially faster than classical computers. Recent demonstrations of quantum advantage [4] confirm the potential of this technology, which simultaneously threatens the security of widely deployed public-key cryptosystems based on integer factorization and discrete logarithms. Shor's algorithm [29] can solve these problems in polynomial time on a quantum computer, necessitating a transition to post-quantum cryptography.

Post-quantum cryptography aims to develop cryptographic primitives secure against both classical and quantum adversaries. Complexity-theoretic evidence suggests that NP-hard problems may resist quantum attacks [8, 13], making them promising foundations for post-quantum security. Among the main approaches, code-based cryptography has emerged as a leading candidate, with several schemes advancing to the final rounds of the NIST post-quantum standardization process [2].

The foundation of code-based cryptography dates to McEliece's 1978 cryptosystem [25], whose security relies on the NP-completeness of syndrome decoding problems [10]. While the original construction using Goppa codes [18, 19] remains unbroken, its practical adoption has been hindered by large key sizes (reaching megabytes). Subsequent variants have sought to improve efficiency by replacing Goppa codes with more structured alternatives [12], though often at the cost of reduced security guarantees [11].

Quasi-cyclic (QC) codes have emerged as a particularly promising family, enabling compact representations that address the key size problem. Notably, two NIST finalists—BIKE [26] and HQC [1]—utilize Quasi-Cyclic (QC) codes, albeit through substantially different constructions. The security of these schemes relies on variants of the Syndrome Decoding Problem tailored to QC codes, known as the Quasi-Cyclic Syndrome Decoding Problem (QC-SDP).

QC codes exhibit desirable cryptographic properties for certain parameters, including minimum weight characteristics and syndrome distributions that resemble those of random codes [14, 16, 24, 17, 15]. This makes them particularly attractive for code-based cryptography. While the cryptographic community generally considers QC-SDP to be hard [28], and the best known attacks only achieve constant-factor improvements over generic syndrome decoding [27, 21], a formal complexity treatment has remained elusive.

Our contribution. This paper establishes that the Quasi-Cyclic Syndrome Decoding Problem (QC-SDP) is NP-hard and that its decision variant is NP-complete (Theorem 3), thereby providing the first formal complexity proof for general QC codes. To achieve this result, we introduce a novel characterization of QC codes (Theorems 1 and 2) that generalizes their structure beyond the circulant matrices typically used in cryptographic applications. This characterization reveals deeper connections between QC codes and random codes, serving as the foundation for our complexity analysis. We further discuss the cryptographic implications of these findings, particularly for NIST post-quantum finalists that rely on QC codes, and examine the practical limitations of our theoretical results in real-world cryptographic deployments.

Related work and problem definition. The computational complexity of quasi-cyclic syndrome decoding was first formally addressed by Berger et al. [9], who provided an explicit definition of QC-SDP. Their analysis established that decoding quasi-cyclic syndromes is NP-hard when the parity-check matrix has the block-diagonal form

$$A = \begin{pmatrix} H & 0 \\ 0 & H \end{pmatrix},$$

where H is the parity-check matrix of an SDP instance. They further claimed NP-completeness for the decision variant. However, this result applies only to a restricted subclass of QC codes and does not extend to the general case.

Subsequent cryptographic designs have relied on the assumed hardness of QC-SDP without formal complexity guarantees. For instance, Aguilar-Melchor et al. [1] explicitly assume the problem's hardness as a foundational hypothesis, citing cryptanalytic evidence from [27, 21] rather than complexity-theoretic proofs. This perspective reflects a broader consensus within the cryptographic community, which has consistently treated QC-SDP as a hard problem in practice [28, 22], despite the absence of formal complexity results. However, existing analyses often restrict attention to specific parameter regimes—such as small index values [28] or systematic parity-check matrices [22]—leaving a general complexity treatment as an open research problem.

2 Preliminaries

Let $\mathbb{F}_q$ be the Galois field of order a prime power q and let $\mathbb{F}_q^n$ denote the n-dimensional vector space defined over $\mathbb{F}_q$.

Definition 1. *A $[n,k]_q$ linear code $\mathcal{C}$ over $\mathbb{F}_q$ of length n and dimension k ($n > k$) is a k-dimensional subspace of $\mathbb{F}_q^n$, which can be represented by two matrices; a $k \times n$ generator matrix G, such that $\mathcal{C} = \{mG,\ m \in \mathbb{F}_q^k\}$ or by a $(n-k) \times n$ parity-check matrix H, such that $\mathcal{C} = \{c \in \mathbb{F}_q^n,\ Hc^T = 0\}$, where $c \in \mathcal{C}$.*

Definition 2. *[30] A Quasi-Cyclic (QC) code of index n_0 is a linear code with dimension $k = p \cdot k_0$, length $n = p \cdot n_0$ and have the property that each cyclic shift of a codeword by n_0 symbols yields another valid codeword.*

Definition 3. *Let $x, y \in \mathcal{C}$. The Hamming distance $d(x,y)$ from x to y, is the number of places at which x and y differ. The Hamming weight $w(x)$ of x is the number of nonzero coordinates in x; i.e., $w(x) = d(x,0)$, where 0 is the zero word. The minimum distance of $\mathcal{C}$, denoted by $d_{\min}$, is $d_{\min}(\mathcal{C}) = \min\{d(x,y) : x \neq y\}$.*

If $\mathcal{C}$ has minimum distance $d_{\min}$, then is an exactly $t = \lfloor (d_{\min} - 1)/2 \rfloor$-error-correcting code.

Definition 4. *Let $\mathcal{C}$ be an $[n,k]_q$ linear code and let H be a parity-check matrix for $\mathcal{C}$. For any $e \in \mathbb{F}_q^n$, the syndrome s of e is the vector $s = He^T \in \mathbb{F}_q^{n-k}$.*

For any syndrome $s \in \mathbb{F}_q^{n-k}$ and parity-check matrix H, we denote by $S_H^{-1}(s) = \{e \in \mathbb{F}_q^n : He^T = s\}$ the set of all vectors in $\mathbb{F}_q^n$ with syndrome s. By definition, $S_H^{-1}(0) = \mathcal{C}$ for any parity-check matrix H of the code $\mathcal{C}$. These syndrome sets correspond precisely to the cosets of the quotient space $\mathbb{F}_q^n/\mathcal{C}$, where each coset $a + \mathcal{C} = \{a + c : c \in \mathcal{C}\}$ contains exactly q^k vectors and represents all vectors sharing the same syndrome. The q^{n-k} distinct cosets form a partition of $\mathbb{F}_q^n$.

Given a received vector $y \in \mathbb{F}_q^n$ with syndrome $s = Hy^T$, the decoding problem consists of finding either a codeword $c \in \mathcal{C}$ that minimizes the Hamming distance $d(y,c)$ (subject to $d(y,c) \leq t$), or equivalently, an error vector $e \in y + \mathcal{C}$ with Hamming weight $w(e) \leq t$. From a computational complexity perspective, the corresponding decision problems are defined as follows:

Definition 5 (**Decisional Syndrome Decoding Problem (D-SDP)**)**.** *Given a random matrix H, a syndrome s and an integer $t > 0$, determine if exist a vector e, with $w(e) \leq t$, such that $s = He^T$.*

Definition 6 (**Decisional Codeword Finding Problem (D-CFP)**)**.** *Given a random matrix H and an integer $t > 0$, determine if exist a vector e, with $w(e) \leq t$, such that $He^T = 0$.*

The NP-completeness of these decisional problems was first established for binary linear codes by Berlekamp, McEliece, and van Tilborg [10]. This result was subsequently extended by Barg to arbitrary finite fields [7], and more recently generalized by Weger et al. to arbitrary finite rings endowed with an additive weight [31]. These progressive generalizations demonstrate the robustness of syndrome decoding complexity across various algebraic structures.

Computational complexity theory establishes that any NP-complete decision problem admits a search-to-decision reduction [3]. This implies that the search variants SDP and CFP are at least as hard as their NP-complete decision counterparts. Although only decision problems formally belong to the NP-complete class, by a widely accepted convention—justified by these polynomial-time reductions—the search problems SDP and CFP are also referred to as NP-complete in the literature.

In the case of QC codes, the definition of the Quasi-Cyclic Syndrome Decoding Problem (QC-SDP) is as follows.

Definition 7. ***[Quasi-Cyclic Syndrome Decoding Problem (QC-SDP)]*** *Given a parity-check matrix H of a QC code, a syndrome s and an integer $t > 0$, find a vector e, with $w(e) \le t$, such that $s = He^T$.*

For our complexity analysis, we consider QC codes with explicit parameters (see Definition 2) that capture the quasi-cyclic structure. Specifically, we work with QC codes of length $n = n_0 \cdot p$ and dimension $k = k_0 \cdot p$, where n_0 is the index. In this parametrized form, QC-SDP can be stated as:

Definition 8 (QC-SDP). *Let $\mathcal{C}$ be an $[n_0 \cdot p, k_0 \cdot p]$ QC code over $\mathbb{F}_2$ with index n_0, $H_{(n_0-k_0)p \times k_0 p}$ being its parity-check matrix, and $s \in \mathbb{F}_2^{(n_0-k_0)p}$ a syndrome vector. Find a vector $e \in \mathbb{F}_2^{k_0 p}$ with weight $w(e) \le t$ such that $He^T = s$.*

3 New characterization of QC codes

QC codes are traditionally defined via circulant matrices. A $p \times p$ matrix C is circulant if it has the form

$$C = \begin{pmatrix} c_0 & c_1 & \cdots & c_{p-1} \\ c_{p-1} & c_0 & \cdots & c_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & \cdots & c_0 \end{pmatrix},$$

where $c_i \in \mathbb{F}_q, 0 \le i \le p-1$. The standard representation of QC code parity-check matrices is given by:

Definition 9. *[6] The parity-check matrix H of a QC code has the form.*

$$H = \begin{pmatrix} H_{00} & H_{01} & \cdots & H_{0(n_0-1)} \\ H_{10} & H_{11} & \cdots & H_{1(n_0-1)} \\ \vdots & \vdots & \ddots & \vdots \\ H_{(r_0-1)0} & H_{(r_0-1)1} & \cdots & H_{(r_0-1)(n_0-1)} \end{pmatrix}$$

where each submatrix $H_{ij}, 0 \le i \le r_0-1, 0 \le j \le n_0-1$, is a circulant matrix of size $p \times p$.

However, a more general construction exists that relaxes the circulant constraint. The following result, established in Lemma 3.2 of [6], shows that QC codes can be defined using arbitrary submatrices arranged in a block-circulant pattern:

Theorem 1. *[6] Let H_i, $0 \le i \le p-1$ be p any matrices, where each H_i has size $r_0 \times n_0$. The parity-check matrix*

$$H = \begin{pmatrix} H_0 & H_1 & \dots & H_{p-1} \\ H_{p-1} & H_0 & \dots & H_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ H_1 & H_2 & \dots & H_0 \end{pmatrix} \tag{1}$$

define a QC code with length $n = p \cdot n_0$ and dimension $k = p \cdot (n_0 - r_0)$.

While Theorem 1 provides a sufficient condition for constructing QC codes, the converse—whether every QC code admits such a representation—has remained open. We now establish this completeness result, thereby obtaining a full characterization of QC codes:

Theorem 2. *If $\mathcal{C}$ is a QC code of length $n = p \cdot n_0$ and dimension $k = p \cdot (n_0 - r_0)$, then the matrix*

$$H = \begin{pmatrix} H_0 & H_1 & \dots & H_{p-1} \\ H_{p-1} & H_0 & \dots & H_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ H_1 & H_2 & \dots & H_0 \end{pmatrix}$$

is a parity-check matrix for $\mathcal{C}$, where each matrix H_i, $0 \le i \le p-1$ has size $r_0 \times n_0$.

Proof. Let H' be a parity-check matrix of a QC code of length $n = p \cdot n_0$ and dimension $k = p \cdot (n_0 - r_0)$. Then H' can be written as follows

$$H' = \begin{pmatrix} H_{00} & H_{01} & \dots & H_{0(p-1)} \\ H_{10} & H_{11} & \dots & H_{1(p-1)} \\ \vdots & \vdots & \ddots & \vdots \\ H_{(p-1)0} & H_{(p-1)1} & \dots & H_{(p-1)(p-1)} \end{pmatrix}$$

where each matrix H_{ij}, $0 \le i, j \le p-1$ has size $r_0 \times n_0$. For $1 \le x \le p-1$, we denote by $c^{(x)}$ the right cyclic shift by $x \cdot n_0$ positions. By the quasi-cyclic property, $c^{(x)} \in \mathcal{C}$ for all $1 \le x \le p-1$, and thus:

$$H' \cdot \left(c^{(x)}\right)^T = 0, \ 1 \le x \le p-1.$$

Now consider the case $x = 1$. The system $H'(c^{(1)})^T = 0$ expands to:

$$\begin{aligned} H_{00}c_{p-1}^T + H_{01}c_0^T + \dots + H_{0(p-1)}c_{p-2}^T &= 0 \\ H_{10}c_{p-1}^T + H_{11}c_0^T + \dots + H_{1(p-1)}c_{p-2}^T &= 0 \\ &\vdots \\ H_{(p-1)0}c_{p-1}^T + H_{(p-1)1}c_0^T + \dots + H_{(p-1)(p-1)}c_{p-2}^T &= 0 \end{aligned}$$

Comparing this with the original system $H'c^T = 0$ we observe that equation 1 in $H' \cdot c^T = 0$ is the same as equation 2 in $H' \cdot \left(c^{(1)}\right)^T = 0$. Since these equalities hold for all codewords $c \in \mathcal{C}$, we obtain the block equalities:

$$H_{00} = H_{11}, \quad H_{01} = H_{12}, \quad \ldots, \quad H_{0(p-1)} = H_{10}$$

Proceeding iteratively for $x = 2, 3, \ldots, p-1$, we obtain the complete set of relations:

$$\begin{gathered} H_{00} = H_{11} = H_{22} = \cdots = H_{(p-1)(p-1)} \\ H_{01} = H_{12} = H_{23} = \cdots = H_{(p-1)0} \\ H_{02} = H_{13} = H_{24} = \cdots = H_{(p-1)1} \\ \vdots \\ H_{0(p-1)} = H_{10} = H_{21} = \cdots = H_{(p-1)(p-2)} \end{gathered}$$

These relations precisely enforce the block-circulant structure of equation (1), completing the proof. □

Theorems 1 and 2 together provide a complete characterization of QC codes: a code is quasi-cyclic if and only if it admits a parity-check matrix in the block-circulant form (1) with arbitrary component matrices. This characterization generalizes the traditional circulant-based definition and will serve as the foundation for our complexity analysis of QC-SDP.

4 NP-completeness of QC-SDP

In this section, we establish the computational complexity of the QC-SDP. Using the novel characterization of QC codes introduced in Section 3, we demonstrate that QC-SDP is NP-hard and that its decision variant is NP-complete.

We now state the main complexity results:

Theorem 3. *The Quasi-Cyclic Syndrome Decoding Problem (QC-SDP) is NP-hard.*

Proof Sketch. Theorems 1 and 2 provide a novel characterization of QC codes via parity-check matrices composed of arbitrary—not necessarily circulant—submatrices. This structural generalization allows us to represent any QC code with given parameters using a block-circulant matrix H. To establish NP-hardness, we construct a polynomial-time reduction from the classical Syndrome Decoding Problem (SDP) to QC-SDP. Given an arbitrary SDP instance, we embed it into a specially crafted QC-SDP instance using the generalized parity-check matrix from Theorem 1. We then show that any efficient algorithm solving this QC-SDP instance would also solve the original SDP instance. Since the decision version of SDP is NP-complete and its search version is NP-hard, it follows that QC-SDP is also NP-hard.

Proof. Let (H, t, s) be an arbitrary instance of SDP, and consider a QC code with length $n = p \cdot n_0$ and dimension $k = p \cdot (n_0 - r_0)$. The QC-SDP instance (H', t', s') is defined as follows. The matrix H' is

$$H' = \begin{pmatrix} H & H_1 & \dots & H_{p-1} \\ H_{p-1} & H & \dots & H_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ H_1 & H_2 & \dots & H \end{pmatrix}$$

where the matrices $H_i,\ 1 \leq i \leq p-1$ are of size $r_0 \times n_0$ and are randomly selected. Let $x \in \mathbb{F}_q^n$ be any vector. Since $n = p \cdot n_0$, x can be seen as $x = (x_0, x_1, \dots, x_{p-1})$ where each $x_i,\ 0 \leq i \leq p-1$ has length n_0. The value of t' is defined as $t + l$, with $l = \sum_{i=1}^{p-1} w(x_i)$. The syndrome s' is defined as $s' = (s + s_0, s_1, \dots, s_{p-1})$ where

$$H_1 \cdot x_1^T + \dots + H_{p-1} \cdot x_{p-1}^T = s_0$$

$$H_{p-1} \cdot x_0^T + H \cdot x_1^T + \dots + H_{p-2} \cdot x_{p-1}^T = s_1$$

$$\vdots$$

$$H_1 \cdot x_0^T + H_2 \cdot x_1^T + \dots + H \cdot x_{p-1}^T = s_{p-1}$$

and each $s_i,\ 0 \leq i \leq p-1$ has length r_0.

Let $\mathcal{A}$ be a polynomial-time algorithm that solves the instance (H', t', s') of the QC-SDP. Then, through $\mathcal{A}$, we can find a vector $e' = (e_1, e_2, \dots, e_n) \in \mathbb{F}_q^n$ with $w(e') \leq t'$ such that

$$H' \cdot e'^T = s'$$

The vector e' can be seen as $e' = (e_0, e_1, \dots, e_{p-1})$ where each $e_i,\ 0 \leq i \leq p-1$ has length n_0. By construction of the QC-SDP instance $\sum_{i=1}^{p-1} w(e_i) = l$. Then, we have that $H' \cdot e'^T = s'$ is

$$H \cdot e_0^T + H_1 \cdot e_1^T + \dots + H_{p-1} \cdot e_{p-1}^T = s + s_0$$

$$H_{p-1} \cdot e_0^T + H \cdot e_1^T + \cdots + H_{p-2} \cdot e_{p-1}^T = s_1$$

$$\vdots$$

$$H_1 \cdot e_0^T + H_2 \cdot e_1^T + \cdots + H \cdot e_{p-1}^T = s_{p-1}$$

By definition, $s_0 = \sum_{i=1}^{p-1} H_i \cdot e_i^T$. Substituting s_0 in the first of the above equations yields

$$H \cdot e_0^T = s$$

Since $w(e) \leq t'$, it follows that $w(e_0) \leq t' - l = t$. Therefore, e_0 is a solution to the SDP instance. Thus, the QC-SDP is NP-hard. □

Theorem 4. *The Decisional Quasi-Cyclic Syndrome Decoding Problem (D-QC-SDP) is NP-complete.*

Proof. Membership in NP is straightforward: given a candidate vector e, one can verify in polynomial time whether $w(e) \leq t$ y $He^T = s$. NP-hardness follows directly from Theorem 3. Therefore, D-QC-SDP is NP-complete. □

5 Cryptographic significance and practical limitations

The demonstration that QC-SDP is NP-complete establishes its worst-case hardness, providing an important theoretical foundation. However, as with many complexity results in cryptography, several important considerations must be addressed. While worst-case hardness is necessary for cryptographic security, it is not sufficient. Cryptographic applications require average-case hardness [23, 20], meaning that random instances of the problem must be hard to solve.

BIKE [26] employs QC-MDPC (Quasi-Cyclic Moderate Density Parity-Check) codes of index 2. Although our general QC-SDP result does not directly apply to the specific parameter regime used in BIKE, the quasi-cyclic structure in BIKE shares fundamental properties with the general QC codes analyzed in our work. The NP-hardness of general QC-SDP suggests that the quasi-cyclic structure alone does not trivialize the syndrome decoding problem. However, we emphasize that BIKE uses codes of index 2 with specific density parameters, and our general result does not preclude the existence of specialized attacks for these particular parameters.

HQC [1] employs a different approach, using a concatenated structure of Reed-Muller and Reed-Solomon codes with quasi-cyclic organization. Our results have more limited direct applicability to HQC because Reed-Muller and Reed-Solomon codes possess rich algebraic structure that may enable specialized decoding algorithms not captured by our general QC-SDP formulation.

The gap between worst-case complexity and cryptographic practice is well-known in code-based cryptography [5]. Future work should investigate whether QC-SDP exhibits average-case hardness for suitable parameter distributions. A significant limitation arises from the fact that our reduction allows arbitrary submatrices H_i, while all practical QC-based cryptosystems use circulant matrices for efficiency. Determining whether QC-SDP remains NP-hard when restricted to circulant matrices is a crucial open problem that directly impacts the theoretical foundations of these schemes.

In conclusion, while our NP-completeness result for QC-SDP provides significant theoretical support for quasi-cyclic code-based cryptography, careful consideration of parameter choices and structural variants remains essential for cryptographic applications.

6 Conclusion

This paper has resolved a fundamental problem in code-based cryptography by formally establishing the computational complexity of the Quasi-Cyclic Syndrome Decoding Problem (QC-SDP). Our main contribution demonstrates that QC-SDP is NP-hard and that its decision variant (D-QC-SDP) is NP-complete, thereby providing the first rigorous complexity characterization for this problem in its general form.

The starting point of our analysis was the development of a novel characterization of quasi-cyclic codes that generalizes their structure beyond the circulant matrices traditionally used in cryptographic applications. This characterization, encapsulated in Theorems 1 and 2, establishes that a code is quasi-cyclic if and only if it admits a parity-check matrix in block-circulant form with arbitrary component submatrices. This theoretical framework not only reveals deeper connections between QC codes and random codes but also provides the necessary structural foundation for our complexity analysis.

From a cryptographic perspective, our results offer a solid theoretical foundation for the use of quasi-cyclic codes in post-quantum encryption schemes. While worst-case hardness does not directly imply cryptographic security, which requires average-case hardness, our work establishes that the quasi-cyclic structure by itself does not trivialize the syndrome decoding problem. This validates the cryptographic community's intuition and provides theoretical justification for schemes like BIKE that rely on the hardness of QC-SDP.

However, our work also identifies important limitations and critical future directions. The gap between arbitrary and circulant matrices represents a significant constraint, since all practical cryptographic schemes use circulant matrices for efficiency. Determining whether QC-SDP remains NP-hard when restricted to circulant matrices remains a crucial open problem. Furthermore, investigating the average-case hardness for cryptographically relevant parameter distributions emerges as an essential direction for future research.

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